Trip-induced spatial linkages:

Evidence from the high-seas fishing industry

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Goal for today [for enviro coffe chat]

1. **Overall "econ sniff test."**
   - Is there a scope for economic intuition/insight?
   - Is the idea interesting enough for the general econ audience?
2. **Advice on sharpening the research question (RQ).**
3. **Minimum viable facts to show before a more deep dive.**
   - Key mechanism I am trying to show: Fishing regions are connected through the trip-chaining behavior of the fishing vessel.

Motivation: A spatially linked commons through fish and vessels

- **A spatially linked commons:** Fishing grounds are linked biologically by fish movement and economically by vessels' trips.
- **Fragmented policy:** With limited global coordination, local closures can reshape entire trips, shifting or reducing fishing pressure at grounds connected by vessel routes.
- **Potential research question:** How do vessels' trip and route choices transmit local conservation policies across economically connected fishing grounds?
  - How does closing one ground reshape vessel routes and fishing effort at connected grounds?
  - How does a ground's position on profitable routes affect visits, holding its productivity fixed?

This project?

- **Data:** Spatial data on vessel-level fishing trips & fish stocks.
- **Empirical setting:** Chinese squid fishing on the high seas off South America.
- **What I find (plan to find):**
  1. Fishing grounds for similar species are spatially dispersed.
  2. Fishing vessels visit multiple grounds per trip.
  3. The effect of local policies may depend on routes connecting fishing grounds.

Literature

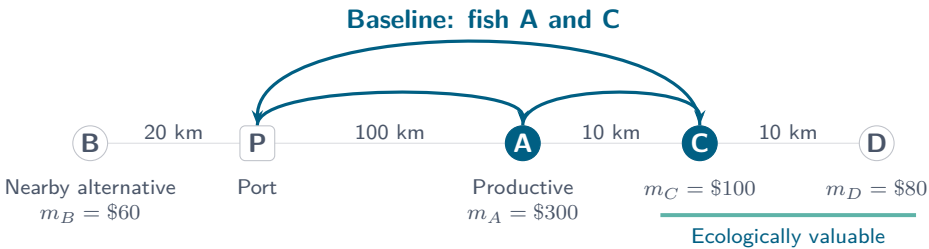
1. Modeling transportation/shipping sector: endogenous trade costs.
   (Hummels et al., 2009; Antràs and Costinot, 2011; Behrens and Picard, 2011; Duranton and Turner, 2011; Bardhan et al., 2013; Allen, 2014; Atkin and Donaldson, 2015; Allen and Atkin, 2016; Ishikawa and Tarui, 2018; Mitra et al., 2018; Allen and Arkolakis, 2019; Bergquist and Dinerstein, 2019; Chatterjee, 2019; Asturias, 2020; Brancaccio et al., 2020; Fajgelbaum and Schaal, 2020; Allen et al., 2022; Grant and Startz, 2022).
   ⇒ Model spatial linkage that arises from agent's trip-chain behavior.

2. Spatial environmental economics: fishers' location choice and fleet dynamics.
   (Bockstael and Opaluch, 1983; Eales and Wilen, 1986; Dupont, 1993; Holland and Sutinen, 1999; Smith, 2000; Curtis and McConnell, 2004; Hutton et al., 2004; Jennings and Lee, 2012; Kroodsma et al., 2018; Dépalle et al., 2021; Wu and Sanchirico, 2026).
   ⇒ Move from single-site location choice to multi-destination trips that link fishing grounds.

3. Theories and policies regarding natural resource (fishery) governance.
   (Gordon, 1954; Clark and Munro, 1975; Sanchirico and Wilen, 1999; Smith and Wilen, 2003; Kellner et al., 2007; Costello et al., 2008; Costello et al., 2016; Sala et al., 2018; Dépalle et al., 2020; Englander et al., 2023; McDonald et al., 2024; Berman, 2026; Berman and Aspelund, n.d.; Berman and Loera, 2026; Loera, 2026).
   ⇒ Evaluate how route-based spatial linkages shape the effects of conservation policies.

Case study: A makes the distant fishing trip worthwhile

A vessel chooses up to **two distinct fishing stops** and returns to port P.



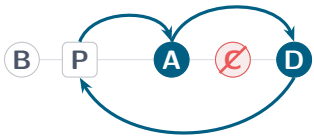
Trip profit: $\underbrace{300 + 100}_{\text{catch margins}} - \underbrace{220 \times 1}_{\text{round-trip fuel cost}} = \mathbf{\$180}$

m_i: revenue less fishing costs per stop. Steaming: \$1/km.

Where we close changes the whole fishing trip

Compare each closure with the baseline A + C trip.

1. Close C



A + D $\pi = \$140$
240 km round trip

Fishing shifts from C to D.

Monitoring: k

2. Close C and D



A + B $\pi = \$120$
240 km round trip

Both valuable grounds protected.

Monitoring: $2k$

3. Close A



B only $\pi = \$20$
40 km round trip

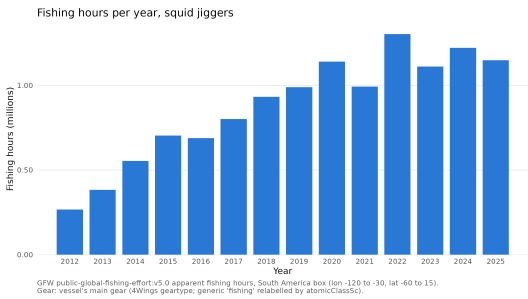
C and D stay open but unfished.

Monitoring: k

k = monitoring cost per closed ground

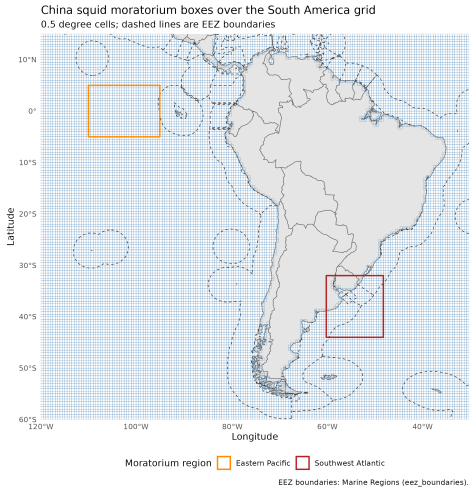
Closing A indirectly protects C and D by changing the vessel's route.

Policy background: High-seas squid fishing in South America



- Squid is a major commercial seafood product (over 3.2 million tonnes caught globally in 2023).
- China's squid-jigging fleet in the Southeast Pacific doubled between 2016 and 2020 ($276 \Rightarrow 557$ vessels).
- In 2017–20, 86% of global squid-fishing effort occurred in waters without squid-stock regulations.

Policy background: Sample region and China squid moratorium



- **Industrial fleet:** In 2019, 516 Chinese squid vessels caught 306,000 tons; 2,500 Peruvian artisanal vessels caught 494,000 tons (Aroni, 2020).
- **Conservation concerns:** Sustainability of shared migratory stocks and competition with local fisheries.
- **Policy response:** Since 2020, China has imposed 3-month annual high-seas moratoria in the Southeast Pacific and Southwest Atlantic.

Data

Vessel trips and fishing activities: Global Fishing Watch

- Vessel tracking data from the Automatic Identification System (AIS) and Vessel Monitoring Systems (VMS) shared by partner countries.
- Machine learning models trained on expert-labeled vessel tracks identify apparent fishing activity from vessel movements. Example
- Vessel-level fishing activity, sequences of port visits, and vessel characteristics, including fishing gear type.

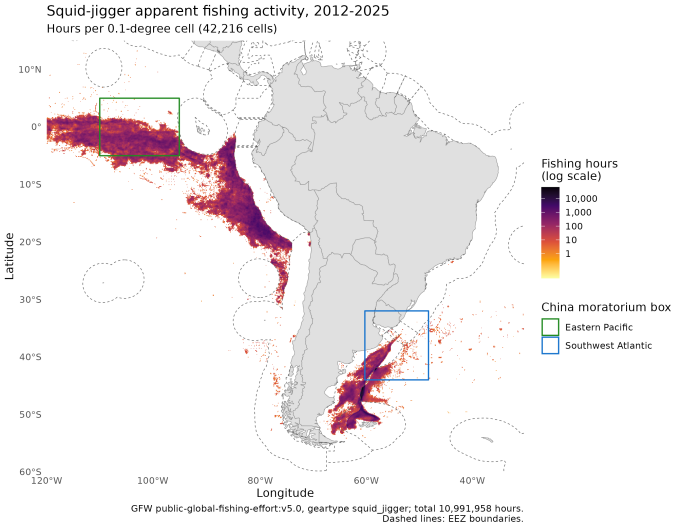
Fish stock/biomass (in progress)

- Stock assessment from catch data? Regional surveys? Production model?

Others

- China's moratorium boundaries and closure dates: Ministry of Agriculture and Rural Affairs (2021).

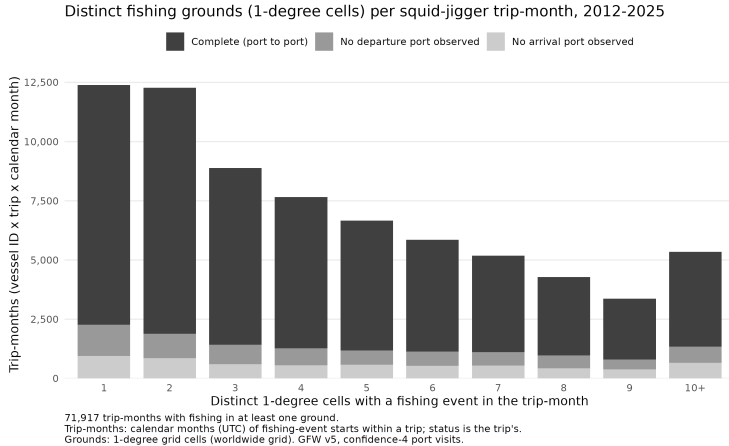
Fact 1: There are multiple fishing grounds!



- Squid fishing grounds are spread across South America.
- This is indirectly evident from fishing activities across space.

▸ CPUE / biomass?

Fact 2: Fishing vessels visit multiple fishing grounds per trip!



Many trips visit multiple fishing grounds.

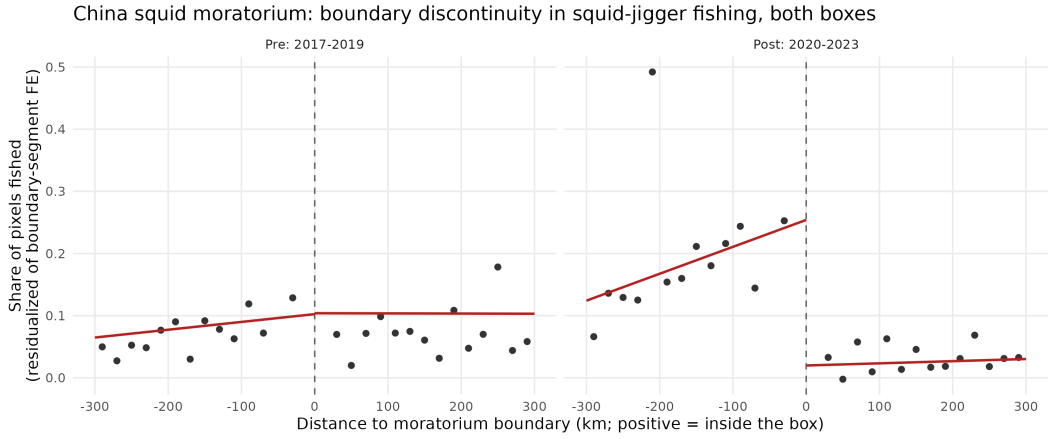
▸ Coarser definition of fishing ground

Fact 3: Closure policy displaces fishing efforts!

$$\text{Fished}_p = \beta\, \mathbf{1}\{d_p > 0\} + f(d_p) + \mu_{s(p)} + \varepsilon_p, \tag{1}$$

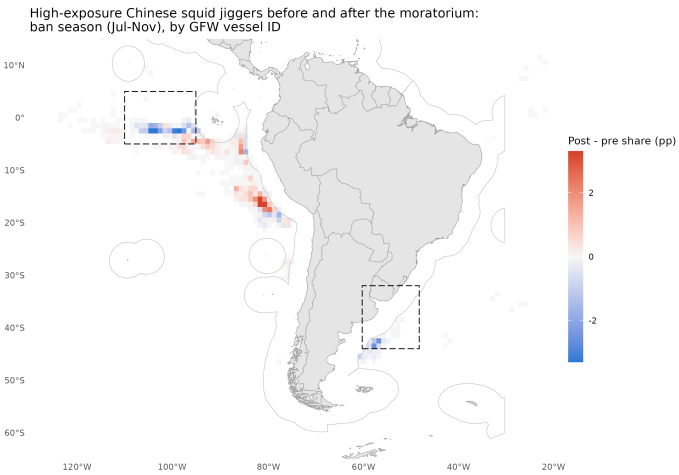
- Spatial RD at the high-seas boundaries of the two China moratorium boxes; p is a 0.5° ocean pixel.
- $\text{Fished}_p = 1$ if p has any squid-jigger fishing during *its own box's ban months*: pre 2017–2019, post 2020–2023.
- d_p: distance (km) to the nearest moratorium boundary, positive inside;
- $\mu_{s(p)}$: box $\times$ 5° boundary-segment FE.
- β: jump in the probability of fishing just inside vs. just outside the boundary. Conley SEs (300 km).

Fact 3: Closure policy displaces fishing efforts!



0.5-degree pixels, EEZ waters removed; moratorium months only (SW Atlantic Jul-Sep, E Pacific Sep-Nov); 4Wings squid_jigger hours.
Points: 20 km bin means within 300 km; lines: local linear fits on each side, estimated on the full 557 km bandwidth. Post includes 2020 (pilot year).

Fact 3: . . . so where are they actually going?



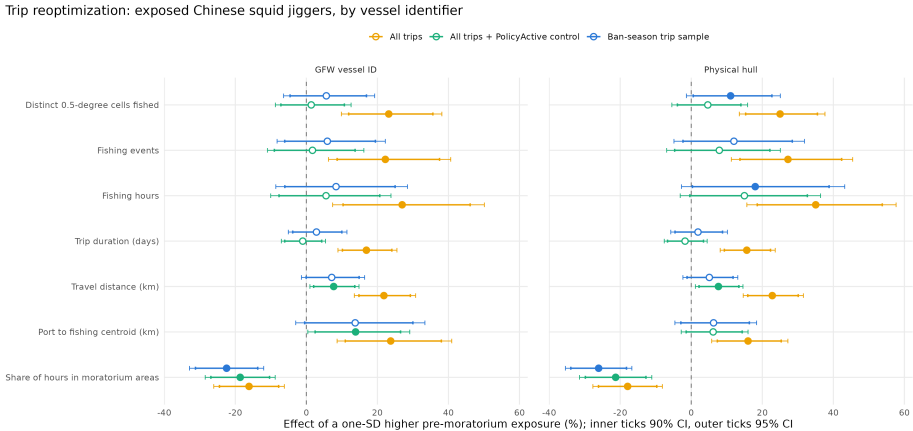
- During the moratorium, their effort shifts to other fishing grounds.
- Potentially to farther areas as well.
- Not just nearby leakage.

Fact 4: Does fishing vessels reoptimize (the whole trip) after the closure?

$$\log y_{im} = \beta\Big(\overbrace{\text{Exposure}_i^{\text{pre}}}^{\text{vessel's exposure to moratorium}} \times \underbrace{\text{PolicyActive}_m}_{\text{trip at sea during a ban season}}\Big) + \delta_i + \delta_{d(m)} + \varepsilon_{im}. \tag{2}$$

- I compare changes in vessel activity per trip before and after the moratorium (i, m is vessel and trip).
- $\text{Exposure}_i^{\text{pre}}$: of i's Jul–Nov fishing hours in 2017–2019, the share inside a moratorium area during its ban months (SW Atlantic Jul–Sep, E Pacific Sep–Nov).
- $\text{PolicyActive}_m = \mathbf{1}\{m \text{ is at sea during a Jul–Nov ban season, 2021–2025}\}$.
- δ_i: vessel FE; $\delta_{d(m)}$: departure year-month FE; SEs clustered by vessel.
- y_{im}: distinct $0.5°$ cells, fishing events and hours, trip days, travel km, port-to-fishing-centroid km (in logs); share of hours in moratorium areas (level).

Fact 4: Does fishing vessels reoptimize (the whole trip) after the closure?



OLS on log(y); share outcome in levels (% of mean). Exposure: share of the vessel's Jul-Nov 2017-2019 fishing hours in moratorium areas during their ban months. Exposure x PolicyActive (trip overlaps Jul-Nov 2021-2025); vessel and departure year-month FE; SEs clustered by vessel. '+ PolicyActive control': PolicyActive also enters on its own. 'Ban-season trip sample': only trips overlapping a Jul-Nov window, 2017-2019 vs 2021-2025. Filled point: significant at 10%; white: not significant. GFW vessel ID: 2,725 trips, 640 vessels (2,123 in the ban-season trip sample) physical hull: 3,639 trips, 559 vessels (2,880 in the ban-season trip sample).

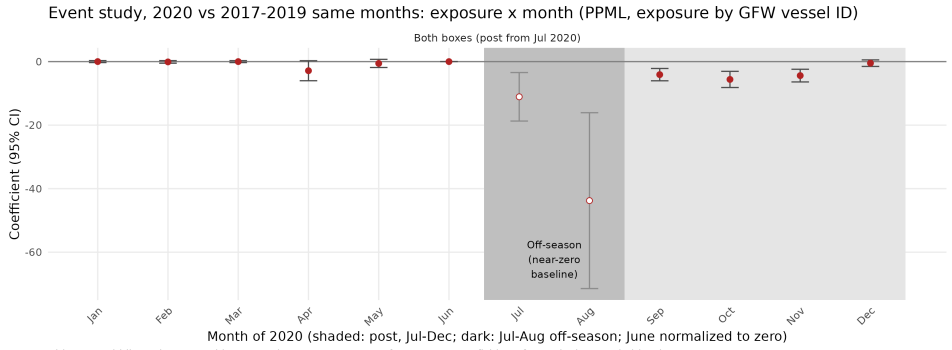
Fact 5: Fishing grounds may be connected through the vessel's route!

$$\text{FishingHours}_{kt}^{CHN} = \sum_{m \neq \text{Jun}} \beta_m \left(E_k \times \mathbf{1}\{t = m, 2020\} \right) + E_k \times \mu_{m(t)} + \delta_k + \delta_t + \varepsilon_{kt} \tag{3}$$

$$E_k = 100 \times \frac{\sum_{v \in V} \text{Hours}_{vk}^{2017\text{–}19}}{\sum_{v \in V} \text{Hours}_{v}^{2017\text{–}19}} \tag{4}$$

- k: $0.5°$ grid outside the moratorium areas; t: year-month (2017–2020).
- $\text{FishingHours}_{kt}^{CHN}$: Chinese squid-jigger hours in k (PPML).
- V: Chinese squid jiggers that fished in a moratorium area in 2017–2019; E_k: share of their fishing in grid k.
- $E_k \times \mu_{m(t)}$: each grid's usual seasonality (calendar month, 2017–2019).
- β_m: Chinese squid jigging in month m of 2020 vs. the same month in 2017–2019, in grids on displaced vessels' routes.

Fact 5: Fishing grounds may be connected through the vessel's route!



Regions connected to the moratorium areas see fishing effort fall.

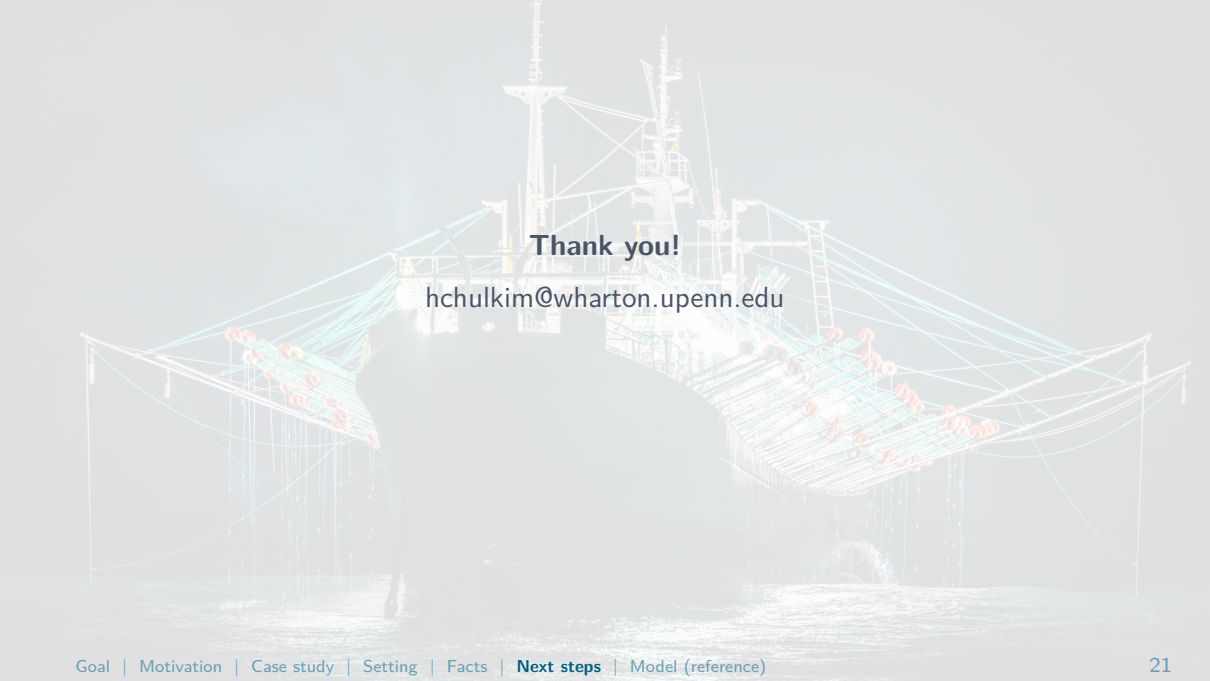
Next steps

Takeaways

1. Vessel trips connect distinct fishing grounds.
2. Spatial closures can shift fishing effort across grounds linked by vessel routes.
3. Fisheries policy design and evaluation may need to account for these spillovers, including across jurisdictions.

Next steps

1. Collect additional fish stock and biomass data (in progress).
2. Develop a formal model of route-based spillovers if supported by the descriptive evidence (do you think it's convincing?).



Thank you!

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7

Model (reference)

The empirical patterns the model aims to explain

Empirical patterns: fishing falls in some open areas after a connected area closes.

1. **Multi-location activity.** Many squid-jigger trip-months include fishing in several grid cells.
2. **Direct policy response.** Boundary comparisons and vessel regressions report reduced fishing use inside moratorium areas.
3. **Spillover response.** The 2020 exposure event study reports lower Chinese fishing hours in historically connected, unclosed cells.

Environment and timing

- Grounds $j = 1, \ldots, J$, a port 0, vessels $i = 1, \ldots, N$, and periods t.
- Ground j has beginning-of-period biomass X_{jt}, price p_j, and catchability q_j.
- A route $r = (0, j_1, \ldots, j_K, 0)$ is an *ordered* list of fishing stops. A vessel selects one route, or stays in port.
- Every visited ground receives fixed effort $\bar{h}$; stop profit is $v_j(X_{jt}) = (p_j q_j X_{jt} - c_j)\bar{h}$.

Within each period: observe stocks/weather/access $\rightarrow$ choose routes and harvest $\rightarrow$ update stocks.

Trip costs

For an operating itinerary r, let $L(r)$ be its directed travel legs:

$$C_{it}(r; w_t) = F_i + \sum_{(k,\ell)\in L(r)} c^{\text{travel}}_{ik\ell,t}(w_t), \qquad C_{it}(\varnothing) = 0. \tag{5}$$

- F_i is the fixed cost of departing; each leg has fuel and travel-time costs.
- Winds, waves and currents (ω_t) alter route costs and potentially feasibility.
- The same offshore journey can serve multiple nearby grounds; the cost of adding a stop can be much less than making a stand-alone trip.

Itinerary complementarities arise from shared fixed and travel costs, even if the grounds' fish stocks are independent.

Vessels choose profitable itineraries

Let $S(r)$ be the set of grounds fished along route r. For $r \neq \varnothing$, a vessel earns

$$V_{it}(r; X_t, w_t) = \sum_{j \in S(r)} (p_j q_j X_{jt} - c_j)\bar{h} - C_{it}(r; w_t), \qquad V_{it}(\varnothing) = 0. \tag{6}$$

Given closures d_t, the vessel solves

$$\boxed{r^*_{it}(X_t, d_t) \in \arg \max_{r \in \mathcal{R}_{it}(d_t, w_t) \cup \{\varnothing\}} V_{it}(r; X_t, w_t).} \tag{7}$$

- $\mathcal{R}_{it}$ contains routes that are legal and fit the voyage-time limit.

Feasible routes

An operating route belongs to $\mathcal{R}_{it}(d_t, w_t)$ if

$$\sum_{(k,\ell)\in L(r)} \tau_{ik\ell,t}(w_t) + |S(r)|\bar{h} \leq T_i, \qquad a_{ijt} = 1 \quad \forall j \in S(r). \tag{8}$$

- Each ground is fished at most once per period. Staying in port is always feasible and yields zero profit.

Effort, harvest, and remaining biomass

The selected routes imply fishing effort at each ground:

$$E_{jt} = \bar{h} \sum_i \mathbf{1}\{j \in S(r_{it}^*)\} + E_{jt}^{\text{other}}. \tag{9}$$

Harvest and post-harvest biomass are

$$H_{jt} = q_j X_{jt} E_{jt}, \qquad S_{jt} = X_{jt} - H_{jt} = X_{jt}(1 - q_j E_{jt}). \tag{10}$$

- Closures affect E_{jt} through vessel route selections.
- Other fleets can also deplete stocks or respond to policies.

Stock dynamics feed back into route profitability

Begin with biologically independent grounds:

$$X_{j,t+1} = R_j(S_{jt}, z_{jt}), \qquad S_{jt} = X_{jt}(1 - q_j E_{jt}). \tag{11}$$

One illustrative increasing, density-dependent stock transition is

$$R_j(S, z) = \frac{a_j(z) S}{1 + b_j S}, \qquad a_j(z) > 0,\ b_j > 0. \tag{12}$$

- z_{jt} captures survival/recruitment conditions.
- Lower harvest leaves more biomass and increases next-period stock under this transition.
- Future $X_{j,t+1}$ changes the stop return $v_j(X_{j,t+1})$ and thus future routes.

Optimality is a comparison across routes

A feasible route r is optimal precisely when

$$V_i(r) \geq V_i(s) \quad \text{for every feasible } s, \quad \text{including } s = \varnothing. \tag{13}$$

For two operating routes r and s, the comparison is

$$\bar{h}\left[\sum_{j \in S(r)\setminus S(s)} (p_j q_j X_j - c_j) - \sum_{j \in S(s)\setminus S(r)} (p_j q_j X_j - c_j)\right] \geq C_i(r) - C_i(s). \tag{14}$$

- Returns at common stops cancel; only the differences in stops and travel costs matter.

A stock threshold for adding a fishing stop

Compare route r with feasible route r^+ that also fishes at B. Define $\Delta C_{iB}(r) = C_i(r^+) - C_i(r)$.

$$V_i(r^+) - V_i(r) = (p_B q_B X_B - c_B)\bar{h} - \Delta C_{iB}(r). \tag{15}$$

Thus, adding B to this trip weakly raises profit iff

$$\boxed{X_B \geq \frac{c_B + \Delta C_{iB}(r)/\bar{h}}{p_B q_B}.} \tag{16}$$

- A shared voyage lowers ΔC_{iB} and the stock threshold for visiting B.
- A closure can eliminate r^+ even when X_B is unchanged.

A companion ground can depend on another stop

Consider grounds A, B and port, with returns v_A, v_B. Let $C(A, B)$ be the cheapest feasible route fishing both grounds. Suppose the joint route is feasible and

$$\boxed{v_A \geq C(A), \qquad C(A, B) - C(A) < v_B < C(B).} \tag{17}$$

Then:

- A alone is weakly profitable: $v_A - C(A) \geq 0$.
- Adding B to the A trip is profitable: $v_B > C(A, B) - C(A)$.
- B alone is unprofitable: $v_B - C(B) < 0$.

Before closure the vessel chooses $A \to B$; after closing A, it stays in port, so open ground B loses effort.

An exact condition for losing an open ground

For policy d, define the best feasible profit *with* and *without* fishing B:

$$M_i^B(d) = \max_{\substack{r \in \mathcal{R}_i(d) \\ B \in S(r)}} V_i(r),$$
$$M_i^{-B}(d) = \max_{\substack{r \in \mathcal{R}_i(d) \cup \{\varnothing\} \\ B \notin S(r)}} V_i(r). \tag{18}$$

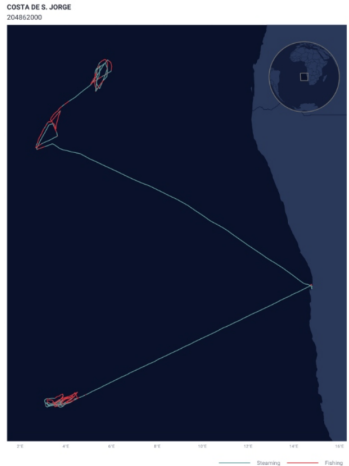
With unique choices, vessel i abandons B after closure of A iff

$$\boxed{M_i^B(0) > M_i^{-B}(0), \qquad M_i^B(d_A) < M_i^{-B}(d_A).} \tag{19}$$

8

Appendix

Example of a fishing activity

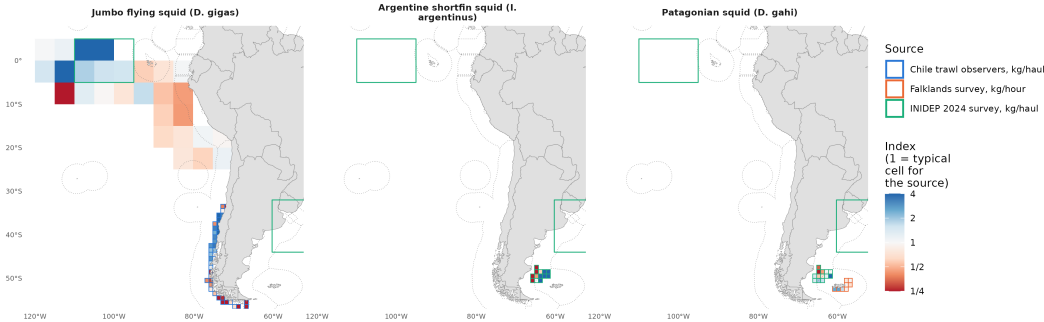


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Catch per unit of effort (Biomass?)

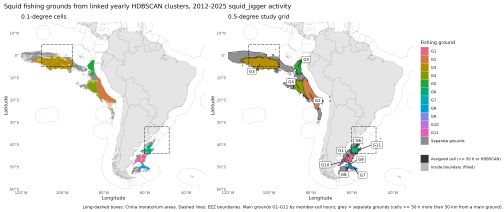
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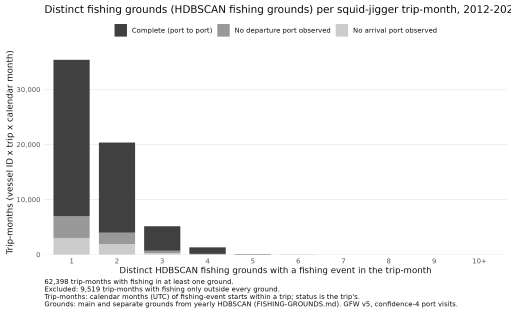
Relative squid abundance index by 1-degree cell, all years pooled



Index: a cell's mean catch rate over all years divided by the median of that source's cell means
(sources differ in gear and units, so values compare places and periods within a source, not across sources).
SPRFMO: Chinese jiggers, 2019-2021 only, 5-degree values copied to 1-degree cells. Chile: trawl hauls with catch, 2010-2019.
Falklands: fixed-station pre-season survey trawls. INIDEP: one Feb 2024 survey. Catch rates are not biomass. Colours capped at 1/4 and 4.
Green boxes: China moratorium areas. Dotted: EEZ boundaries.

Coarser definition of fishing ground





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